

12345678

UNNI C02 SMARTIFICATION — REV A

Two-sheet cleanup: standard power symbols, grid-aligned wiring, explicit local connectivity.

USB + Power

ESP32—C6 + Auxiliary

Simulation Harness

File: 01_usb_power.kicad_sch

File: 02_esp_aux.kicad_sch

File: 03_simulation_harness.kicad_sch

Unni C02 Smartification

Sheet: /
File: unni-smartification-c6.kicad_sch

Title: Unni C02 Smartification — ESP32—C6 Rev A

Size: A3Date: 2026-08-19

KiCad E.D.A. 10.0.5Rev: A—v28—finalId: 1/4

A

B

C

D

E

F

1

2

3

4

5

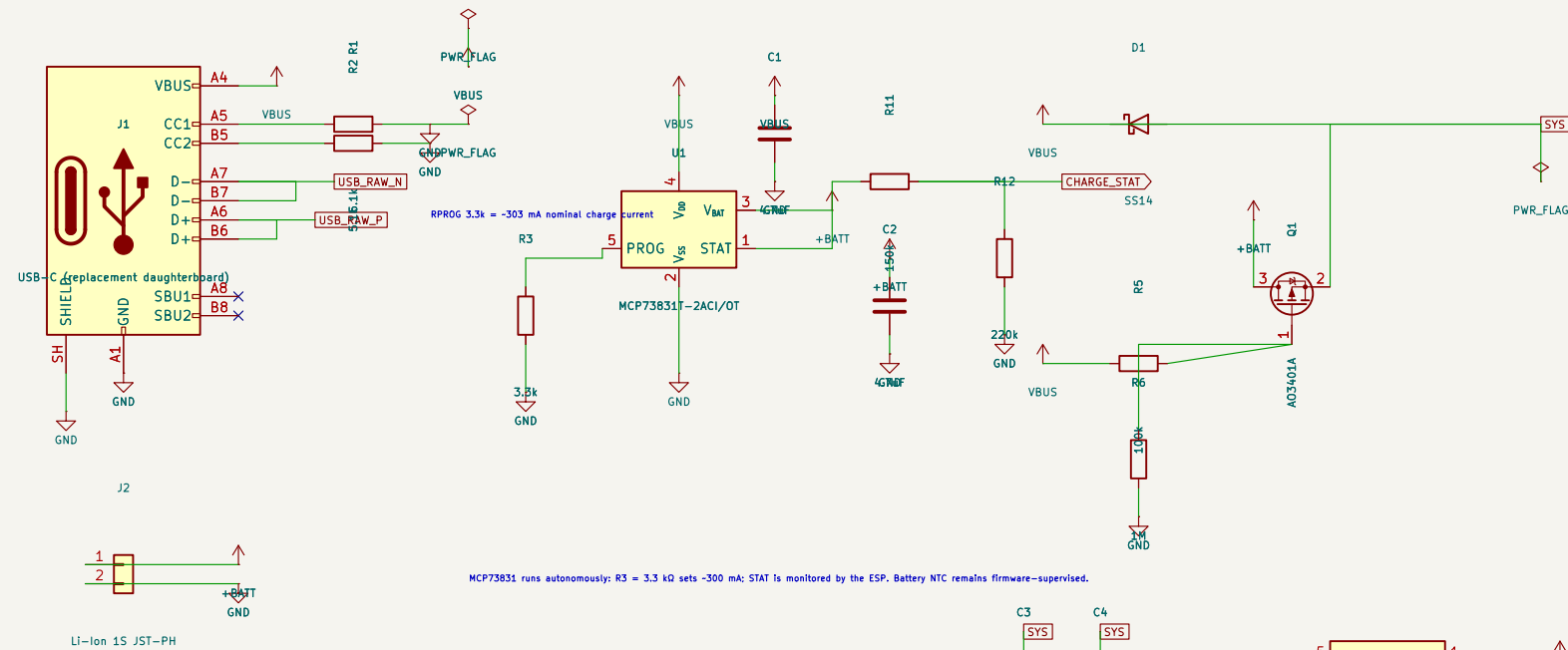
6

7

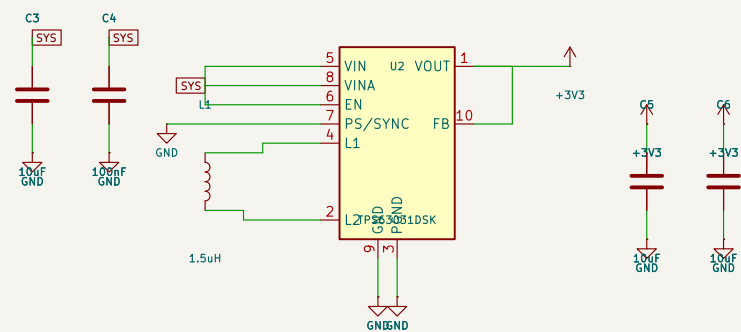
8

USB + POWER

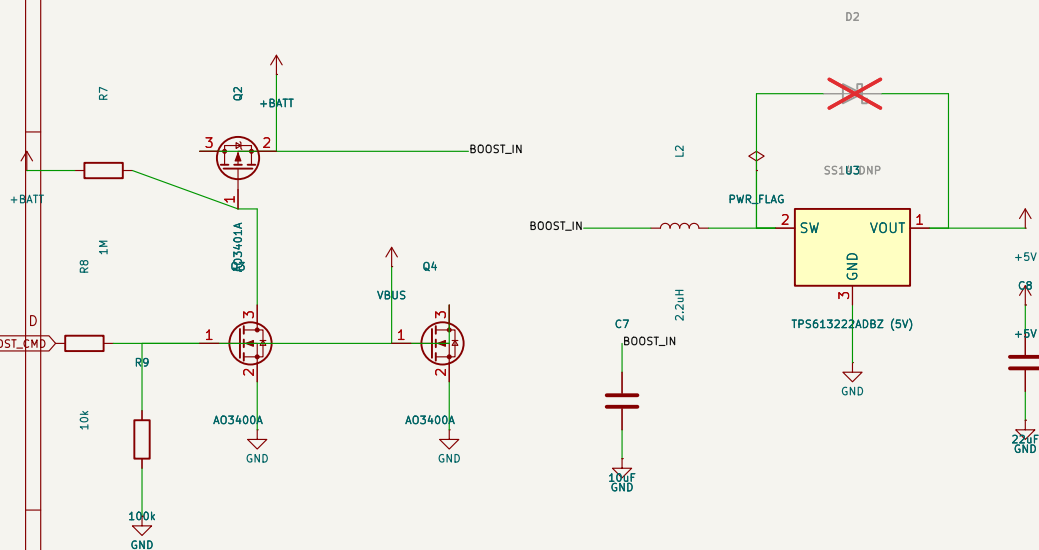
All placement/wiring is on 50 mil (1.27 mm) grid. Standard KiCad symbols used where available.



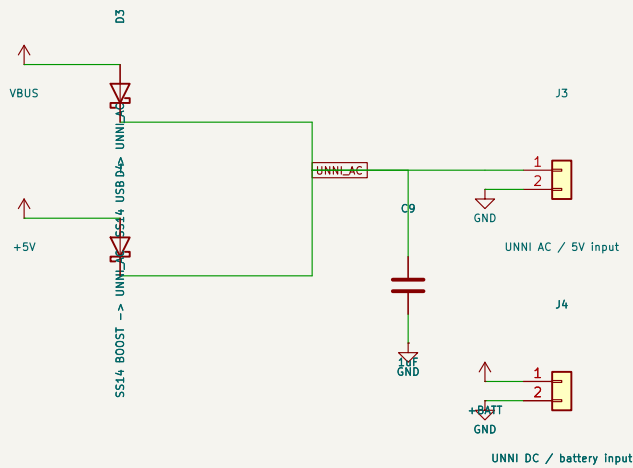
MCP73831 runs autonomously; R3 = 3.3 kΩ sets ~300 mA; STAT is monitored by the ESP. Battery NTC remains firmware-supervised.



FORCED-AWAKE 5V



UNNI POWER OUTPUTS



Unni C02 Smartification

Sheet: /USB + Power/
File: 01_usb_power.kicad_sch

Title: USB + Power

Size: A3

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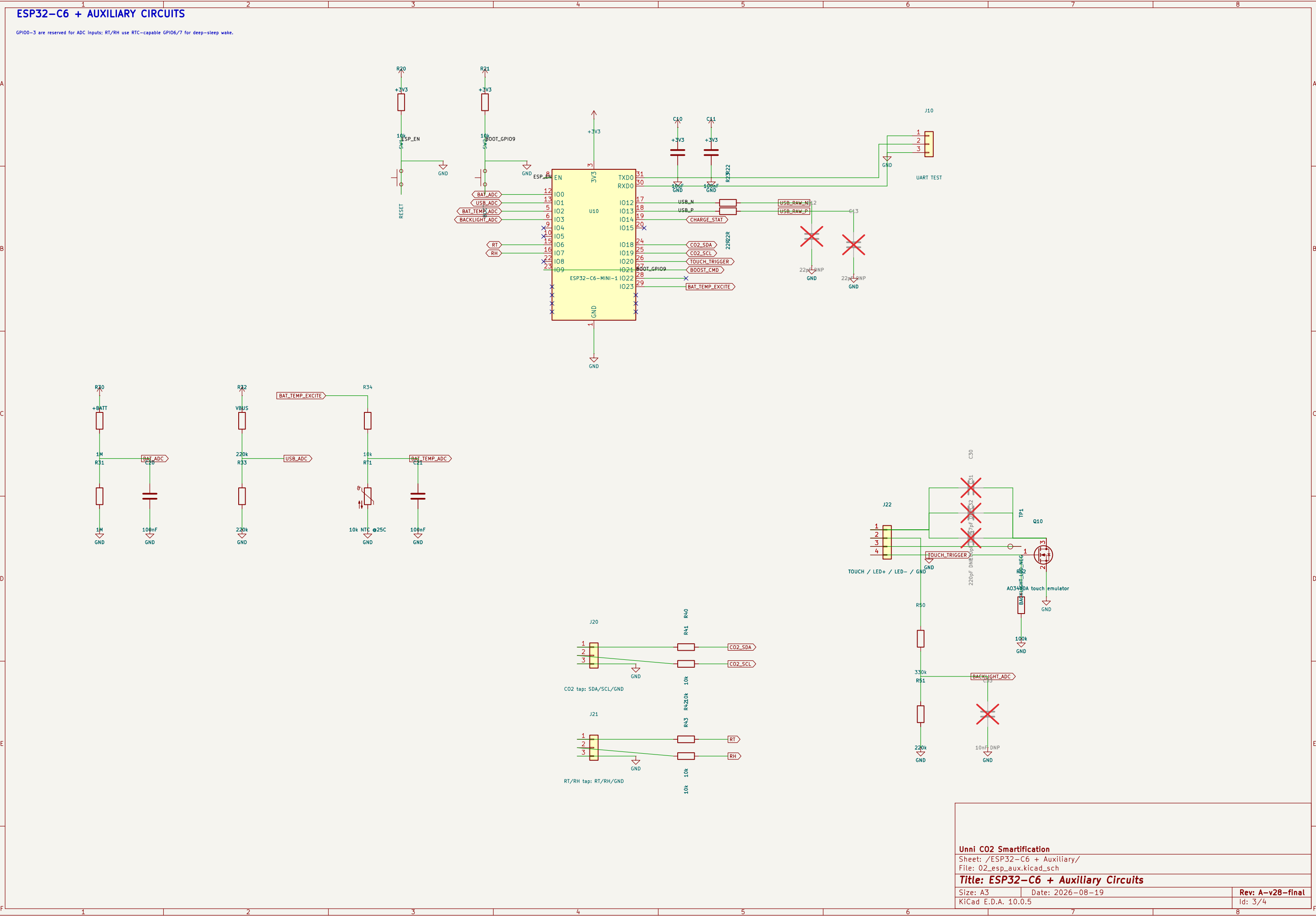
Rev: A-v28-final

KiCad E.D.A. 10.0.5

Id: 2/4

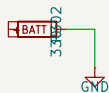
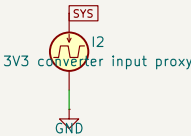
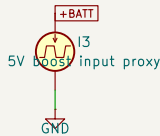
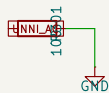
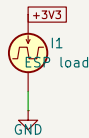
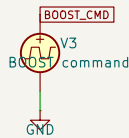
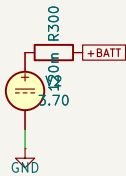
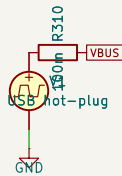
ESP32-C6 + AUXILIARY CIRCUITS

GPIO0-3 are reserved for ADC inputs; RT/RH use RTC-capable GPIO6/7 for deep-sleep wake.



SIMULATION HARNESS – excluded from BOM/PCB

Scenario: battery-only → forced awake → USB hot-plug while BOOST_CMD remains asserted → autonomous charging → USB removal. Battery model includes 120 mΩ ESR. I2/I3 proxy converter input power so battery sag is visible without switch-level regulator models.



Unni C02 Smartification
Sheet: /Simulation Harness/
File: 03_simulation_harness.kicad_sch

Title: Simulation Harness

Size: A3 Date: 2026-08-19
KiCad E.D.A. 10.0.5

Rev: A-v28-final
Id: 4/4